

MTH166

Lecture-27

**Boundary Value Problems on Wave
Equations**

Topic:

Solution of Partial Differential Equations

Learning Outcomes:

To solve Boundary value problems of one dimensional Wave Equation.

Recall: Wave Equation

Equation: $\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2}$

Nature: Hyperbolic

Solution: 1. $u(x, t) = (ax + b)(ct + d)$

2. $u(x, t) = (ae^{px} + be^{-px})(ce^{Cpt} + de^{-Cpt})$

or

$$u(x, t) = (a \cosh px + b \sinh px)(c \cosh Cpt + d \sinh Cpt)$$

3. $u(x, t) = XT = (a \cos px + b \sin px)(c \cos Cpt + d \sin Cpt)$ **(Most suitable one)**

Problem. Solve $\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2}$, $0 \leq x \leq l, t > 0$ under boundary conditions:

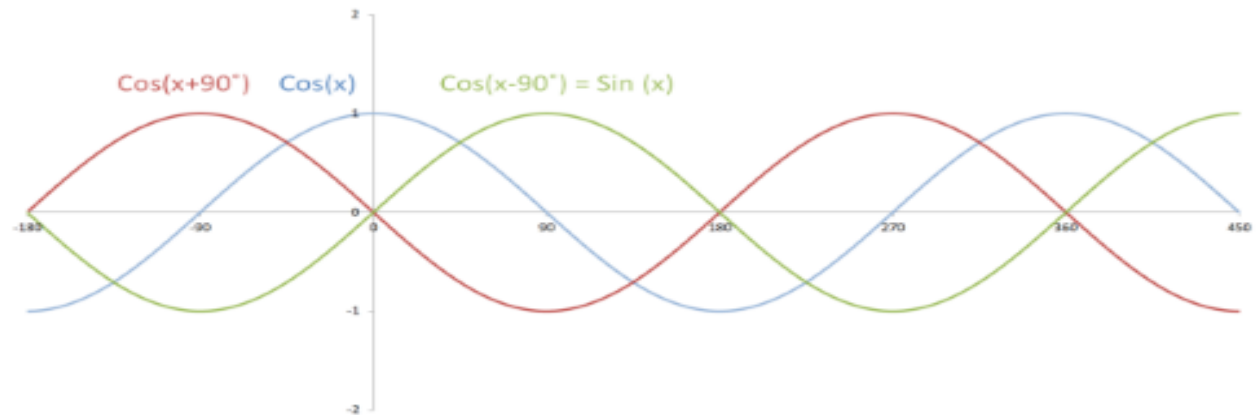
$$u(0, t) = 0, u(l, t) = 0.$$

Solution. The given one dimensional wave equation is:

$$\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2} \quad (1)$$

$$0 \leq x \leq l \text{ (length)}, t > 0 \text{ (time)}$$

$$\text{Such that : } u(0, t) = 0, u(l, t) = 0.$$



We know the most appropriate solution of equation (1) is:

$$u(x, t) = (a \cos px + b \sin px) (c \cos Cpt + d \sin Cpt) \quad (2)$$

Using the condition $u(0, t) = 0$, we get:

$$u(0, t) = (a \cos 0 + b \sin 0) (c \cos Cpt + d \sin Cpt)$$

$$\Rightarrow 0 = a(c \cos Cpt + d \sin Cpt) \quad \Rightarrow \mathbf{a = 0}$$

Equation (2) becomes:

$$u(x, t) = (b \sin px)(c \cos Cpt + d \sin Cpt) \quad (3)$$

Using the condition $u(l, t) = 0$, we get:

$$u(l, t) = (b \sin pl)(c \cos Cpt + d \sin Cpt)$$

$$\Rightarrow 0 = (b \sin pl)(c \cos Cpt + d \sin Cpt)$$

$$\Rightarrow 0 = (b \sin pl)(c \cos Cpt + d \sin Cpt)$$

$$\Rightarrow \sin pl = 0 \quad (\text{As } b \neq 0)$$

$$\Rightarrow \sin pl = \sin n\pi$$

$$\Rightarrow pl = n\pi \quad \Rightarrow \mathbf{p} = \frac{n\pi}{l}$$

Equation (3) becomes:

$$u(x, t) = (b \sin px)(c \cos Cpt + d \sin Cpt)$$

$$\Rightarrow u(x, t) = (b \sin \frac{n\pi x}{l})(c \cos \frac{n\pi C}{l} t + d \sin \frac{n\pi C}{l} t)$$

$$\Rightarrow u(x, t) = \left(A \cos \frac{n\pi C}{l} t + B \sin \frac{n\pi C}{l} t \right) \sin \frac{n\pi x}{l} \quad (4)$$

By principle of superposition:

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi C}{l} t + B_n \sin \frac{n\pi C}{l} t \right) \sin \frac{n\pi x}{l} \quad (5)$$

Which is the required solution of Wave equation (1).

Where $A_n = \frac{1}{2l} \int_0^l u(x, 0) \sin \frac{n\pi x}{l} dx$; $u(x, 0)$ is the initial displacement.

And $B_n = \frac{1}{2\pi C} \int_0^l \frac{\partial u}{\partial t}(x, 0) \sin \frac{n\pi x}{l} dx$; $\frac{\partial u}{\partial t}(x, 0)$ is the initial velocity.

Polling Quiz

If $u(x, t) = (a \cos px + b \sin px) (c \cos Cpt + d \sin Cpt)$ is the solution of one-dimensional wave equation, then which of the following is true when initial condition $u(l, t) = 0$ is applied to the solution:

- (A) $a = 0$
- (B) $p = n\pi/l$
- (C) $b = 0$
- (D) $C = 0$

